Class Activity and Notes Sheet: Tropical Cyclone Structure

Learning goals. By the end of class, you should be able to identify the major structural regions of a mature tropical cyclone; distinguish its primary and secondary circulations; describe the radial and vertical structure of tangential, radial, and vertical wind; and explain the warm/dry eye, moist eyewall, eye inversion, and warm-core structure.

1. A tropical cyclone as a vortex

In your own words, distinguish a **wave** from a **vortex**. What does it mean to say that a vortex "carries fluid with it"?

Quick check. Circle the component that dominates the primary circulation of a mature TC:

radial wind tangential wind vertical wind.

2. Primary versus secondary circulation

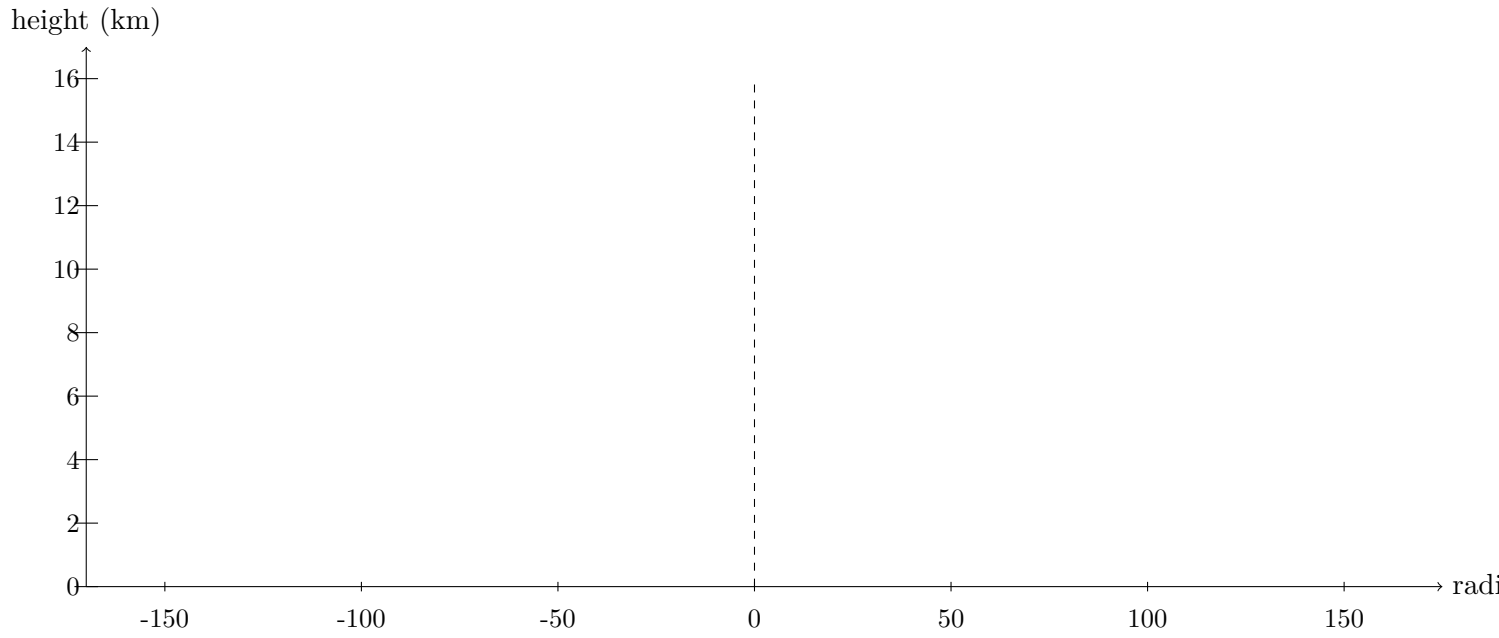
Complete the table during lecture.

	Primary circulation	**Secondary circulation**
Dominant motion		
Low levels		
Eyewall		
Upper troposphere		
Key physical role		

Concept question. Why does surface friction produce radial inflow rather than simply slowing the tangential wind?

3. Build the radius–height circulation

On the schematic below, add and label: (a) eye, (b) eyewall, (c) RMW, (d) boundary-layer inflow, (e) eyewall ascent, (f) upper-level outflow, and (g) eye subsidence. Use arrows to show the secondary circulation.

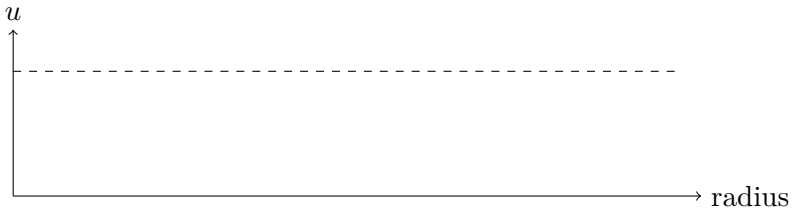


Think-pair-share. The eyewall often slopes outward with height. What does this imply about the radius of strong ascent and strong tangential wind as height increases?

4. Radial structure of the wind

Sketch idealized radial profiles at low levels. Mark the eye, RMW, eyewall, and outer core.

Tangential wind



Radial wind

Complete the statements:

a) Tangential wind increases rapidly outward through the _____ and peaks near the _____.

b) Outside the RMW, tangential wind generally _____ with radius.

c) With increasing height, the maximum tangential wind generally becomes _____ and shifts _____.

d) Low-level radial inflow is strongest in approximately the lowest _____ km.

5. Boundary layer and eyewall ascent

Construct a causal chain using the lecture concepts:

surface stress → _____ → cross-isobaric inflow → _____ → strong ascent

Near the eyewall the boundary-layer flow can become _____. Explain why rapid deceleration there favors convergence and ascent near or inside the RMW.

6. The hurricane eye

Record the expected properties of the mature eye relative to the eyewall/environment.

Property	**Eye**	**Physical interpretation**
Surface pressure		
Free-tropospheric temperature		
Free-tropospheric humidity		
Vertical motion		
Low-level humidity/cloud		
Inversion		

Key Point. Latent heating is concentrated in the eyewall, yet aircraft observations can show the warmest air in the eye. Explain this apparent contradiction.

7. Reading an eye sounding

The lecture's Hurricane Hugo eye sounding shows warm, very dry air above roughly 700 hPa and cooler, nearly saturated air below an inversion. Answer:

a) What process is indicated by the warm, dry layer aloft?

b) Why can the lowest part of the eye remain moist despite subsidence aloft?

c) Draw a horizontal line on a conceptual sounding at the inversion and annotate the air masses above and below it.

8. Interpreting observed cross sections

Use the Hurricane Hilda and composite cross sections shown in lecture to complete these observations.

1. The primary circulation is strongest just above the _____.
2. Below about 500 hPa, the tangential circulation has relatively _____ vertical shear.
3. Near the tropopause far from the center, the circulation can become _____.
4. The largest warm anomaly in the eye is found primarily in the _____ troposphere.
5. Mean radial flow is predominantly _____ at low levels and _____ at upper levels.
6. Mean tangential flow is _____ through a deep layer and can become _____ aloft.

Interpretation. Why is a warm-core temperature anomaly dynamically consistent with the strong cyclonic circulation of a tropical cyclone?

9. Rainbands and secondary eyewalls

From the Hurricane Gilbert example, identify the sequence moving outward from the center:

Eye → _____ → _____ → secondary convection/eyewall → outer rainbands

What is a **moat**? What change in the wind profile might signal a secondary eyewall?

10. Synthesis: tell the story of a mature hurricane

In 5–7 sentences, describe a radial transect from the outer core through the eyewall into the eye. Your explanation must use all of these terms correctly: *inflow, angular momentum, RMW, ascent, outflow, subsidence, warm core, inversion.*

Minute reflection and conceptual question: A student says: "The eye is warm because latent heat is released there." Is this statement correct? Give a physically precise correction in 2–3 sentences.